

Supplementary Material: Simulation Software and Implementation Workflow

This supplementary material documents the software workflow used to construct PhyIRSTD and provides an implementation-level explanation of the three stages in Fig. 3 of the main manuscript. The figures illustrate scene reconstruction, target geometry and material assignment, trajectory configuration, thermal calculation, atmospheric and sensor configuration, and final infrared image generation. The purpose is to clarify how the physical scene and imaging parameters are established before rendering.

The **Complex Environment Infrared Scene Generation Software (XD CAE-IR 1.0)** is an independently developed infrared simulation and analysis platform with proprietary intellectual property rights by Key Laboratory of Optoelectronic Information Perception in Complex Environment, Xidian University. It adopts a unified framework for processing multidisciplinary and heterogeneous data and integrates structural, atmospheric, thermal, and infrared physical models to establish a complete pre-processing, solver, and post-processing workflow. Through efficient three-dimensional scene modeling and physics-based radiative simulation, the software supports quantitative analysis of complex environments, target infrared characteristics, infrared radiation and stealth design, and dynamic infrared scene rendering. It can generate both infrared and visible-light imagery under diverse scene, target, environmental, and sensor configurations.

S1. Background Acquisition and 3D Environment Construction

This section corresponds to Step I of Fig. 3 in the main manuscript. The objective is to reconstruct an observation environment with consistent geographical geometry and material attributes, which can subsequently participate in thermal and radiative calculations.

S1.1. Database access and geographical background selection

The workflow begins by opening the project and resource database, as illustrated in Fig. S1(1). The database provides access to the geographical scene, terrain, model, material, and other resources required for simulation. A geographical background is then selected in the global scene interface, where the user specifies the region of interest and its spatial extent. The selected region provides the geographical reference for the subsequent reconstruction and determines the coordinate framework in which the background geometry, targets, and camera will be placed. The example in Fig. S1(2) shows the selection of the GIS region.

S1.2. Land-cover classification and background material assignment

After selecting the geographical region, the background imagery is processed into a land-cover classification map, as shown in Fig. S1(3). Different surface categories are distinguished so that roads, runways, vegetation, buildings, and other regions can be assigned their corresponding material descriptions. The classification result is inspected and edited in the material editor, as shown in Fig. S1(4). Optical and thermal attributes are associated with the relevant surface classes, including material-dependent radiative properties and thermal boundary settings. Together with terrain elevation and 3D mesh information, these data form the reconstructed background environment used by the subsequent thermal and imaging solvers.

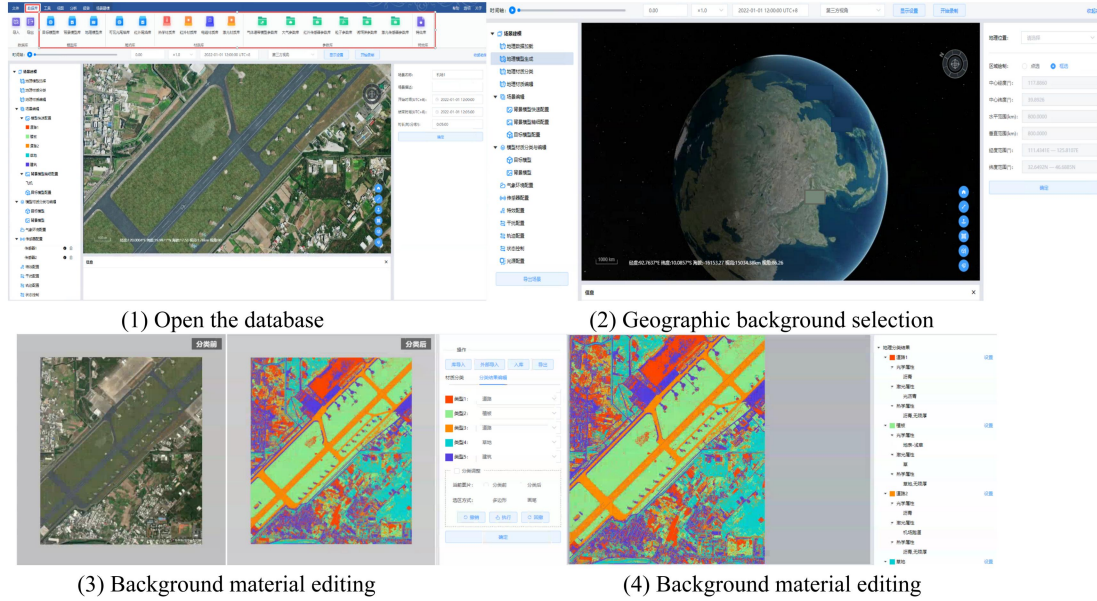


Fig. S1. Background acquisition and material configuration corresponding to Step I of Fig. 3. (1) Opening the resource database. (2) Selecting the geographical background. (3) Land-cover classification and material editing. (4) Assignment and inspection of background material properties.

S2. Target Collection and Physical Target Modeling

This section corresponds to Step II of Fig. 3. It describes the construction of three-dimensional targets, the assignment of surface materials, and the generation of their motion trajectories. The target geometry and the reconstructed background are defined in a common three-dimensional world coordinate system before infrared imaging.

S2.1. Target geometry and material classification

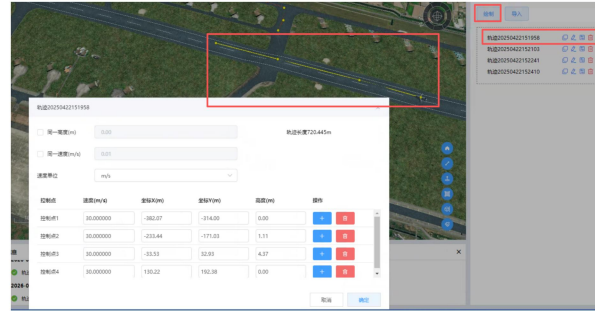
Representative target models are imported into the target library and constructed with reference to the physical dimensions and geometric proportions of their corresponding real-world categories or specific models. Fig. S2(a) illustrates an A320 aircraft model, its surface mesh, and the associated texture and material classification maps. The mesh preserves the geometric structure of the target, while the material map divides its surface into regions to which different optical and thermal properties can be assigned. The material editor associates these regions with the corresponding material definitions, rather than treating the entire target as a single uniform-intensity template. This representation provides the geometric and material basis for the thermal calculation and subsequent radiance generation.

S2.2. Three-dimensional trajectory generation

Target trajectories are configured in the reconstructed scene using three-dimensional control points and motion parameters, as illustrated in Fig. S2(b). The trajectory editor allows the user to define the spatial coordinates, height, and speed of the trajectory segments and to inspect the resulting path in the scene. During sequence generation, the target position and attitude are updated according to the configured motion state. Because the target and background share the same world coordinate system, their relative spatial relationship is established before imaging. The projected target size and visible surface area are then determined by the target geometry, camera position, viewing distance, and attitude, rather than by an independently selected image-space scale.



(5) Target Model Diagram and 3D Material Classification Editing



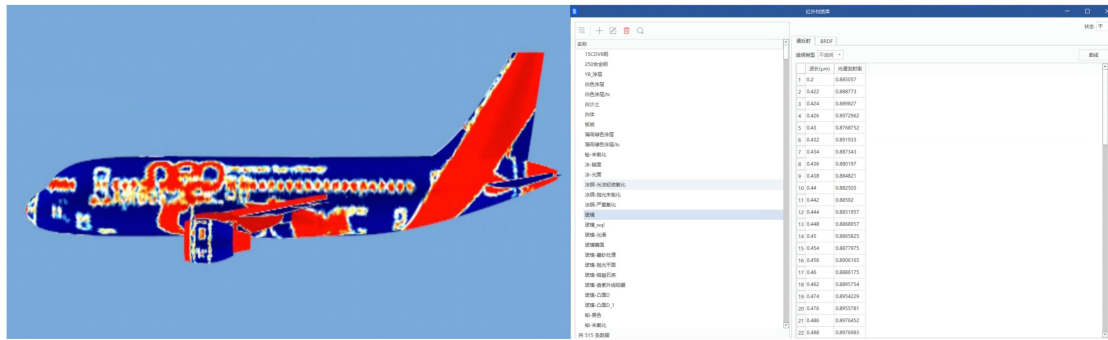
(6) Target Trajectory Generation and Editing

Fig. S2. Target collection and motion configuration corresponding to Step II of Fig. 3. (5) Target model, 3D surface mesh, texture/material classification, and material assignment interface. (6) Three-dimensional trajectory generation and editing.

S2.3. Target thermal calculation and material library

The target material library provides the radiative properties used for infrared simulation. Fig. S3 shows the aircraft thermal-result visualization and a material-library interface containing wavelength-dependent optical-property data. The target mesh is divided into material regions, and the corresponding material properties are associated with the thermal and radiative calculations. The physical material settings are configured with reference to SE-Workbench. In the main manuscript, the target temperature is modeled using one-dimensional heat transfer along the surface-normal direction, with thermal conductivity, density, specific heat capacity, emissivity, absorptivity, and reflectance as material-dependent inputs.

The surface-temperature calculation is initialized from the prescribed thermal state and environmental conditions. Internal heat conduction is coupled with the surface heat exchange, including convection, absorbed incident radiation, and self-emission. The resulting surface temperature is supplied to the gray-body radiation model to calculate the target spectral radiance. Thus, the thermal field provides a physical input to image formation rather than an arbitrary grayscale target template. The colored aircraft image in Fig. S3 is a visualization of the thermal calculation; it is not an absolute-temperature scale and does not by itself establish numerical temperature values. For exact numerical reproduction, the material record must also specify the initial temperature, thermal-layer discretization, time step, inner boundary condition, and any prescribed internal heat source or operating-state condition. These values must be taken from the actual thermal-solver configuration; they cannot be reconstructed from the screenshot alone.



(7) Target Thermal Calculation and Material Library

Fig. S3. Target thermal calculation and material library corresponding to Step II of Fig. 3. The left panel visualizes the aircraft thermal field, and the right panel shows the material database.

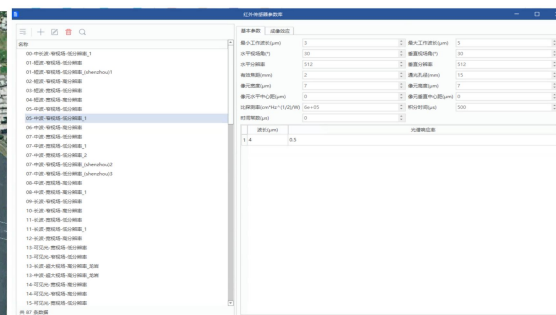
S3. Infrared Physical Modeling and Sensor Imaging

This section corresponds to Step III of Fig. 3. After the background and targets have been configured, the software performs joint scene-level infrared imaging. The scene geometry, thermal/radiative properties, atmospheric propagation, camera projection, and detector response are combined to generate the final infrared frames.

S3.1. Scene assembly and sensor configuration

Fig. S4(8) shows the assembled airport scene with the configured target trajectory. The target, background, and camera are positioned in the same scene, allowing their geometry and visibility to be evaluated consistently. The sensor database is then used to select or configure the imaging system, as shown in Fig. S4(9). The interface provides the spectral working band, field of view, focal-plane resolution, focal length, pupil diameter, detector dimensions, integration time, and spectral response settings. These parameters determine how the scene is projected and sampled by the focal-plane array (FPA).

The example displayed in Fig. S4(9) uses a 3-5 μm working band, a 512×512 detector array, a 30×30 field of view, a 2 mm effective focal length, a 15 mm entrance-pupil diameter, a 7 μm detector size, and a 500 μs integration time. It is noted that these are values visible in the example sensor profile, not a claim that every PhyIRSTD sequence uses this configuration.



(8) Scene modeling

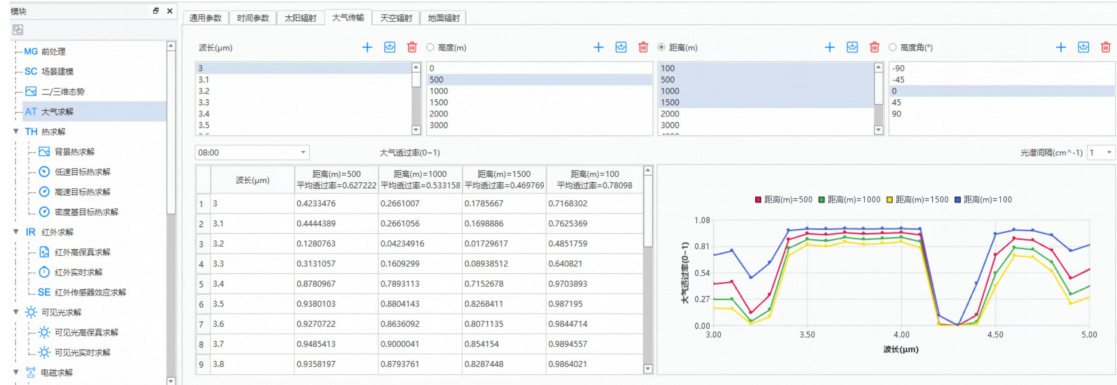
(9) Sensor Parameter Configuration

Fig. S4. Scene assembly and sensor parameter configuration corresponding to Step III of Fig. 3. (8) Reconstructed scene and target trajectory. (9) Sensor database and example spectral, optical, detector, and integration-time settings.

S3.2. Atmospheric transmission and environmental radiation

The atmospheric solver is used to establish the propagation conditions between the scene and the sensor. Fig. S5 shows the configuration of wavelength, altitude, propagation distance, elevation angle, and time, together with calculated spectral transmittance curves. The example interface samples the 3-5 μm band and compares atmospheric transmission at several propagation distances. The resulting curves demonstrate that attenuation is both wavelength- and path-dependent. They therefore provide a more appropriate basis for infrared image formation than a single wavelength-independent attenuation factor.

The atmospheric calculation supplies the transmittance and path-radiance information used in the scene-level radiative-transfer model. The environmental radiation terms further account for sky, solar, ground, and surrounding-background contributions. The background and target surfaces use their assigned thermal and radiative properties, while the atmospheric conditions determine the propagation-dependent attenuation and path contribution. The numerical implementation follows the radiative-transfer and sensor-imaging equations described in Section III-B of the main manuscript, where MODTRAN-derived atmospheric data are used, the corresponding model version, atmospheric profile, gas/aerosol settings, and exported spectral quantities should be recorded with the simulation configuration.



(10) Atmospheric Parameter Calculation

Fig. S5. Atmospheric parameter calculation corresponding to Step III of Fig. 3. The interface shows spectral, altitude, distance, angle, and time settings, together with wavelength-dependent transmittance values and curves for different propagation distances.

S3.3. Sensor effects and high-fidelity infrared image generation

The ideal radiometric image is further processed by the sensor-effect module, as illustrated in Fig. S6(11). The interface provides configurable effects such as low-pass filtering, detector non-uniformity, Gaussian noise, salt-and-pepper noise, and other sensor-related options. The displayed example has low-pass filtering and Gaussian noise enabled, whereas non-uniformity and salt-and-pepper noise are disabled. The visible low-pass filter level is 2 and the Gaussian-noise level is 1. The non-uniformity coefficient of 0.01 and salt-and-pepper noise density of 0.01% are displayed in the interface. The exact meaning and units of these software-specific controls should be documented from the solver configuration before they are interpreted as physical noise variances or detector calibration values.

Fig. S6(12) illustrates the final high-fidelity infrared imaging calculation. The software takes the assembled scene, configured sensor, target thermal state, and atmospheric/environmental parameters as inputs. Radiance is evaluated along detector-directed rays, and the resulting spectral contributions are integrated over the working band using the sensor response. The focal-plane signal is then converted to digital intensity and processed according to the selected sensor-effect configuration. This corresponds to the radiance integration, FPA sampling, quantization, and degradation stages in Fig. 3 of the main

manuscript.

The software also provides options for retaining intermediate calculation results and selecting output information. These intermediate radiance and geometry outputs can be used to inspect the simulation before producing the final infrared sequence. The screenshots demonstrate the existence and configuration of these stages; they do not by themselves establish a measured agreement with a particular real camera.



(11) Sensor Effect Calculation



(12) High-Fidelity Infrared Image Calculation

Fig. S6. Sensor-effect configuration and final infrared image generation corresponding to Step III of Fig. 3. (11) Sensor-effect controls and example infrared result. (12) High-fidelity infrared imaging interface, scene preview, and calculation/output settings.

S4. Annotation Generation

In addition to the radiometric image, each target instance is assigned a unique identifier during scene construction. The target geometry and its identifier are propagated through the same camera projection used for image formation. The projected visible silhouette is recorded to obtain the corresponding pixel-level instance mask, and the bounding box is derived from the minimum and maximum foreground coordinates of that mask. This procedure preserves geometric alignment between the target and its annotations without requiring independent manual contour delineation. The segmentation mask is obtained from the geometric instance representation rather than by thresholding the thermally blurred infrared image. The bounding box annotations follow the VOC format.